Due: August 20, 2026

1. Find all antiderivatives for the following functions:

    a) x^5
    b) $9x^3 - x + 3$
    c) x
    d) 1
    e) $\sqrt{x}$
    f) $\frac{1}{x^{10}}$
    g) $\frac{5}{x}$

2. For the following exercises, find the antiderivative of the function for the provided endpoints, and compute the area under the curve on that interval using Riemann sums:

    a) $f(x) = x^2;\ 1 \leq x \leq 3,\ n = 4$
    b) $f(x) = x^3;\ 1 \leq x \leq 3,\ n = 5$
    c) $f(x) = x^3;\ 0 \leq x \leq 1,\ n = 5$

3. Calculate the following definite integrals:

    a) $\int_1^1 x dx$
    b) $\int_1^2 5 dx$
    c) $\int_1^2 8x^3 dx$
    d) $\int_0^1 4e^{-3x} dx$
    e) $\int_1^4 3\sqrt{x} dx$
    f) $\int_0^5 e^{-2x} dx$
    g) $\int_3^6 x^{-1} dx$
    h) $\int_0^3 (x^3 + x - 7) dx$

4. **R Introduction Questions:**

    a) Explain the difference between an R script (‘.R‘ file) and an R Markdown (‘.Rmd‘ file) document. When would you prefer to use one over the other in your political science research workflow?
    b) What is an Integrated Development Environment (IDE)? Name the standard IDE used for R programming and list two main panels/windows in it and what they show.